

Exact Sampling from Neural Networks: Depth, Attention, and Decoding

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Abstract

A random output can require less information than the value of a function. We study this difference for neural networks, counting input probes, random bits, arithmetic, and precision refinement. A single bounded tanh row admits an exact sample using a constant expected number of input coordinates. When every absolute row sum is at most one, the zero-bias worst-case query cost lies between the square root of depth and depth. The sharp square-root law holds at every fixed first-layer rank and for a bounded column envelope, with sufficient input size.

The main theorem concerns decoding after prefill. If each head’s keys lie in a common affine space of fixed dimension, an exact decoder with positive RMSNorm stabilizers can construct and append its own keys and values in expected time polynomial in model size and polylogarithmic in context length. The proof maintains finite Taylor moments and prepares higher precision in the background. Rare complete refinements preserve the exact original autoregressive law. The decoder’s polynomial degree grows with rank. These bounds give no useful speed prediction at the large ranks of ordinary models. Without a context index, standard score scaling can force linear search; arbitrary growing-dimensional indexed queries also face a conditional hardness barrier. We give checkpoint quantities that determine which positive bound can be applied, and state the unrestricted zero-bias exponent as the main open problem.

1 Why one sample can be cheaper

Computing an average of n arbitrary signs requires reading every sign. Sampling one uniform coordinate gives a sign with that average as its expectation. The same distinction persists through some nonlinear computations. It does not follow by independently replacing every activation with an unbiased random estimate: a nonlinear map of that estimate generally has a different expectation.

We follow the information available to a decoder. A fresh input permits only weight preprocessing. A supplied cache permits an index over keys and values. A complete decoder must also create the next cache records, and must do so as precision increases. These are different algorithmic problems. An input-probe lower bound before prefill does not give a query lower bound after an index has read the context.

This version gives complete proofs of the maintained decoder theorem and its finite-cache building blocks. The full version [17] contains the depth-dependent factory and lower-bound proofs, the sharp critical subclass theorems, and the additive-residual laws. Those results locate the decoder theorem within the broader sampling question. The dominant restriction is geometric: a rank bound independent of width is needed for a uniform polynomial in width. Fixing every parameter

of a full-rank model also gives a formal polylogarithmic context bound, but its exponent can be enormous. This is an asymptotic existence result, with no claimed checkpoint speedup.

2 Model and principal theorem

A dyadic number is an integer divided by a power of two. All fixed parameters have binary encoding length at most b , with denominators dividing 2^b . Let n be residual width, D the number of blocks, V the vocabulary size, $T_0 \geq 1$ the prompt length, and T the current context length. We use one head of dimension n . Rectangular heads are treated in the full version [17]. The norm of a matrix is its maximum absolute row sum, $s(W) = \max_i \sum_j |W_{ij}|$. Embeddings, affine bias coordinates, and all matrix row norms are bounded by $s \geq 1$. Table 1 collects the notation.

For $\varepsilon > 0$, define the unscaled normalization

$$N_\varepsilon(h) = \frac{h}{\sqrt{\varepsilon + \|h\|_2^2/n}}.$$

Each block computes affine queries q_t , keys k_t , and values v_t from $N_\varepsilon(h_t)$ and uses the causal attention mean

$$a_{tj} = \frac{\beta}{n} \langle q_t, k_j \rangle, \quad u_t = \frac{\sum_{j \leq t} e^{a_{tj}} v_j}{\sum_{j \leq t} e^{a_{tj}}}. \quad (2.1)$$

An affine map of u_t is added to h_t . A second normalization, affine map, and coordinatewise tanh supply the other additive update. Either update may be multiplied by a fixed coefficient in $[0, 1]$; convex residual mixtures are allowed as well. An affine head gives V logits z_i , followed by probabilities $p_i = e^{z_i} / \sum_j e^{z_j}$. No bounded-logit promise is needed. All stabilizers satisfy $\varepsilon \geq 2^{-b}$. The temperature is a nonnegative dyadic number or the exact algebraic number $\beta = \sqrt{n}$. Each affine key map is fixed across positions and its linear part has rank at most r . Thus all keys of one layer lie in one known affine space, including keys not yet generated. A separate rank bound at each position would not suffice.

Theorem 2.1 (An exact maintained decoder). *For every fixed r , the preceding model has a uniform exact sampler and a constructive polynomial F_r such that prefill costs*

$$T_0 F_r(n, D, V, b, s, \beta, \log(T_0 + 2)),$$

and the conditional expected bit work of each subsequent token is at most

$$F_r(n, D, V, b, s, \beta, \log(T + 2)).$$

The cost includes constructing and appending its neural keys and values, all random bits, and every precision refinement. The conditional law is the original model’s law for every realized token history. At standard scaling and polynomially bounded D, V, b, s , the cost is $\text{poly}_r(n, \log T)$. The degree depends on r .

The theorem concerns the real functions defined by the stored finite parameters. It does not assert equality with a particular floating-point implementation. It includes pre-norm additive residuals and a positive normalization stabilizer; neither a small row norm nor contraction is needed. Large norms instead increase the moment degree and working precision. The proof occupies Sections 5–7. Figure 1 shows what each kind of preprocessing changes.

Quantity	Meaning	Quantity	Meaning
n, D, V	Width, depth, vocabulary	T, T_0	Current and prompt lengths
b, B	Parameter and total bit scales	r	Common affine key dimension
$Q_{\max}, K$	Query and key coordinate bounds	β	Score temperature
A_{coin}, H	Score-coin amplitude; moment coefficient bound	P, m, N_m	Working bits, Taylor degree, feature count
ε, λ	Stabilizer; certified squared-denominator floor	Q	Distinct fresh-input probes

Table 1: Notation. $B = 16 + b + \lceil \log_2(T + n + D + V + r + 2) \rceil$; fixed rational chart lengths are included when used.

3 A scalar example and the preprocessing boundary

A sign interface for $u/R \in [-1, 1]$ returns independent signs of that mean. Products of independent signs multiply means; random selection among interfaces takes a convex combination. A Bernoulli factory uses such requests to transform an unknown mean. These principles are classical [6, 8, 13, 19].

Proposition 3.1 (One bounded tanh row). *Let $u = c + \sum_j w_j x_j$, where $x_j \in \{-1, 1\}$, $|c| \leq 1$, and $\sum_j |w_j| \leq 1$. After $O(nB^2)$ weight preprocessing, one can return a sign of mean $\tanh u$ using $O(1)$ expected input requests and $O(B^4)$ expected bit work.*

Proof idea. A random coordinate supplies the linear mean. An exponential acceptance race then selects between the two signs with the logistic probabilities.

Proof. Put $A = |c| + \sum_j |w_j| \leq 2$. If $A = 0$, return a fair sign. Otherwise a weighted choice among $\text{sgn}(c)$ and $\text{sgn}(w_j x_j)$ returns a sign of mean u/A . Integer cumulative weights implement this choice exactly in $O(B^2)$ expected bit work after preprocessing. Propose $Y = \pm 1$ uniformly and draw $N \sim \text{Pois}(2A)$. Accept if all N independent bits of bias $(1 + Yu/A)/2$ are one. The Poisson generating function gives acceptance e^{Yu-A} . Hence the accepted sign has mean

$$\frac{e^u - e^{-u}}{e^u + e^{-u}} = \tanh u.$$

A trial succeeds with probability $e^{-A} \cosh u \geq e^{-A}$. Its expected source count is at most $2A$, so renewal gives at most $2Ae^A \leq 4e^2$ requests. Here is a finite-bit Poisson implementation, within the classical Buffon-machine framework [7]. For $\lambda = 2A \leq 4$, propose $K \geq 0$ with probability $2^{-(K+1)}$ using fair bits, and accept with probability $e^{-\lambda} (2\lambda)^K / (128K!)$. This is at most $e^4/128 < 1$. Its accepted mass is $e^{-\lambda} \lambda^K / (256K!)$, so each trial succeeds with probability $1/256$ and the accepted K is Poisson. The rational factor is below $e^8/128 < 52$ and has $O(K[B + \log(K+2)])$ bits. A finite exponential series on $[0, 4]$ gives certified acceptance intervals with polynomial work. Geometric proposal and uniform-comparison tails have all fixed moments; summing their polynomial costs gives $O(B^4)$ expected work. Thus neither the Poisson draw nor the acceptance test uses a real oracle. $\square$

For attention, suppose $|a_j| \leq A_{\text{coin}}$ and a sign of mean a_j/A_{coin} is available. The Poisson test accepts a uniform proposal j with probability $e^{a_j - A_{\text{coin}}}$, giving the exact attention law.

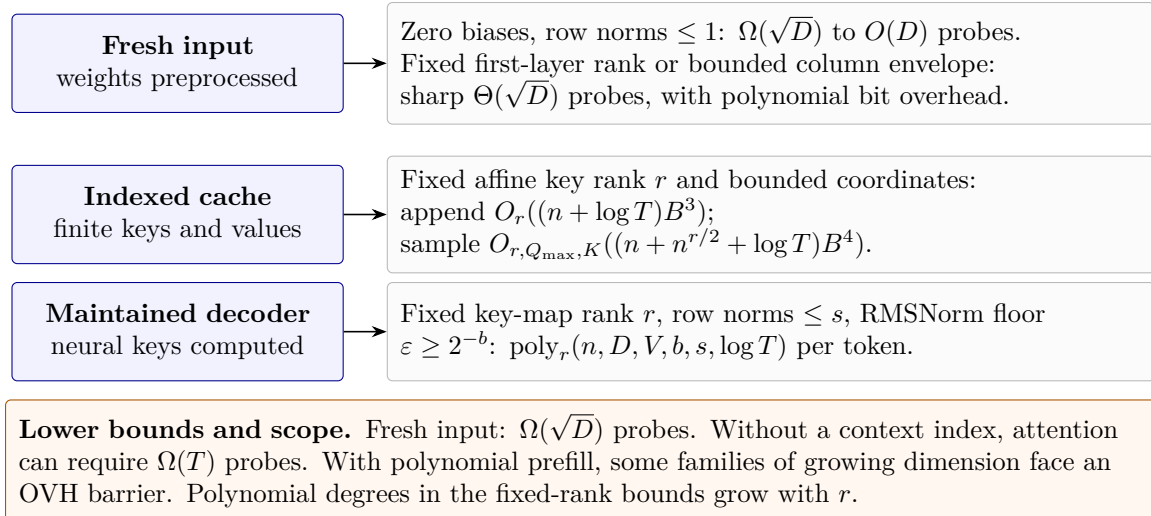


Figure 1: What preprocessing changes. The first row counts expected input probes; the other rows count expected bit work. Every law is exact. Here n is width, D depth, $T \geq 2$ cache length, and V vocabulary; B includes parameter encodings and address bits. Attention uses $\beta = \sqrt{n}$. Finite-cache coordinate bounds $Q_{\max}, K$ are fixed; the decoder incorporates them through n, s . Fresh-input preprocessing uses $D = \lceil \log_2 n \rceil$ and $\lceil \log_2(D + 1) \rceil \leq b = O(\log n)$, where b is parameter precision. Its column envelope is $\sum_j \max_i |w_{1,ij}|$. OVH is the randomized Orthogonal Vectors Hypothesis.

The expected predecessor count, including a fresh value sign, is at most $2e^{2A_{\text{coin}}} - 1$ when each score test uses two predecessor signs; the stopped-test calculation is given in the full version. A bounded score range therefore can remove the linear context factor before prefill. Standard scaling alone cannot.

Here is the search obstruction in its simplest form. At a fresh all-negative sign context, give position j score Ax_j and value x_j . Changing only coordinate i makes its positive attention mass

$$\frac{e^{2A}}{T - 1 + e^{2A}}.$$

Couple an exact sampler's random tapes on the two inputs. Their outputs can differ only after coordinate i is read, so the probability of that read is at least the total-variation distance of the target laws. Summing over i forces $\Omega(\min\{T, e^{2A}\})$ probes. A bounded tanh and binary-softmax head preserve the gap up to an absolute constant; a convex residual gate multiplies it by its weight α . The matching one-block family has cost $\Theta(1 + \alpha \min\{T, e^{2\beta s^2}\})$ [17]. Thus $2\sqrt{n}s^2 \geq \log T$ can force linear context search. An index built from those signs is already different on the two inputs, so this coupling does not yield an online lower bound after prefill.

4 An index for supplied finite keys

A bounded coordinate chart turns fixed affine dimension into a finite number of score cells. The index in this section returns an attention index; Section 5 instead maintains attention means for the complete decoder.

Lemma 4.1 (Bounded rational coordinates). *For an $n \times d$ full-column-rank dyadic matrix U , with $d \leq r$ and b -bit entries, one can find d row indices I so that $A = U(U_I)^{-1}$ satisfies $A_I = I_d$ and $|A_{ij}| \leq 2$. The work is $O_r(nB^3)$ and all chart coefficients have $O_r(B)$ bits.*

Proof idea. A large reconstruction coefficient identifies a row replacement that more than doubles a nonzero integer determinant.

Proof. Clear the common dyadic denominator and start with any nonsingular row minor of the resulting integer matrix E . If $|[E(E_I)^{-1}]_{ij}| > 2$, replace selected row j by row i . Determinant multilinearity multiplies the determinant's absolute value by that coefficient. The new row is distinct because selected rows have identity coefficients. A nonzero integer minor has magnitude at least one; every minor has magnitude at most $d!\|E\|_{\max}^d$. Only $O_r(B)$ replacements occur. Each stage uses $O_r(nB^2)$ schoolbook bit work. Cramer's rule gives the stated coefficient lengths and a common determinant denominator. This is the standard maximum-volume argument [18]. $\square$

Theorem 4.2 (Appendable finite cache). *Fix r , dyadic bounds $K, Q_{\max} > 0$, and $\beta \geq 0$ dyadic or $\sqrt{n}$. A stream of b -bit keys in $[-K, K]^n$, whose every prefix has affine dimension at most r , admits an exact attention-index sampler for explicitly supplied dyadic queries in $[-Q_{\max}, Q_{\max}]^n$. Queries occur at nonempty prefixes. All scalar encodings of keys, queries, coordinate bounds, and dyadic temperature have at most b bits. With $\beta \leq \bar{\beta} \leq 2\beta$ dyadic when $\beta > 0$, its append cost is $O_r((n + \log(T + 2))B^3)$ and its expected query cost is*

$$O_r((n + M_T + \log(T + 2))B^4), \quad M_T \leq \min\{T, (r + 1)(5 + 32r\bar{\beta}Q_{\max}K)^r\}.$$

It stores $O_r((T + 1)nB)$ bits. No append scans previous records.

Proof idea. Keep each old chart when the span grows. Within a chart, a grid makes one score representative a constant-factor envelope for its whole cell.

Proof. The case $\beta = 0$ is uniform sampling. Otherwise fix the first key c . Maintain a basis of differences $k_j - c$. At each increase of rank start a new chart from Lemma 4.1, and retain every earlier chart and its records. At most $r + 1$ charts are ever used. Within one chart, $k = c_0 + Ak_I$, where $c_0 = c - Ac_I$ and $k_I \in [-K, K]^d$. A query's coordinate coefficients obey $|(q^T A)_i| \leq 2nQ_{\max}$. Use grid width $\Delta = K2^{-h}$, with $h = \lceil \log_2 \max(1, 8r\bar{\beta}Q_{\max}K) \rceil$. Two keys in a cell have score difference below $2\beta rQ_{\max}\Delta \leq 1/4$. There are at most $(5 + 32r\bar{\beta}Q_{\max}K)^d$ cells in a chart.

Store occupied cells in a balanced tree and their member identifiers in binary append trees. Membership in the current span is tested by the exact equality $k - c = A(k_I - c_I)$. A failed test creates the next chart; all old records stay in place. An insertion takes the stated work, including a chart change, and identifier storage is linear in T .

At query time compute $q^T A$ and $\langle q, c_0 \rangle$ once per chart. Let a_C be each occupied cell's representative score and $a_0 = \max_C a_C$. Put $w_j = e^{a_j - a_0 - 1/4}$ and $e_C = e^{a_C - a_0}$. Then $w_j \leq e_C < 2w_j$ in that cell, while $Z = \sum_j w_j > 1/2$. Set $p = \lceil \log_2(64T) \rceil$ and compute positive dyadic upper bounds

$$e_C \leq u_C \leq e_C + 3 \cdot 2^{-p}, \quad u_C \geq 2^{-p}.$$

Enclose e_C to width 2^{-p-2} , round the upper endpoint up to the 2^{-p} grid, and add one grid unit. The envelope mass satisfies $Z \leq W = \sum_C |C|u_C < 2Z + 3/64 < (17/8)Z$. Choose a cell with weight $|C|u_C$, then a uniform member j , and accept with probability w_j/u_C . The accepted mass

is exactly w_j/W , so the result has the target law and the expected number of proposals is below $17/8$.

All representative comparisons are rational before multiplication by β/n . At standard scaling an exponent is a rational plus a rational multiple of $1/\sqrt{n}$. Bisection encloses that square root. For a nonpositive exponent, values below $-(p+2)\log 2$ may be enclosed by $[0, 2^{-p-2}]$; on the remaining interval the exponential Taylor series with range reduction costs a fixed polynomial in $B+p$. Near the cutoff either safe enclosure can be used after a constant overlap, so no exact equality test is needed. Since $u_C \geq 2^{-p}$, $t + O(p)$ exponential bits suffice for acceptance at precision t . The probability of an undecided uniform comparison is $O(2^{-t})$. Summing that geometric tail against the polynomial refinement work gives $O(B^4)$ expected work per proposal. Exact integer cumulative weights give the cell choice. One projection per chart and one initial enclosure per occupied cell prove the query bound. All statements condition on the supplied records and query, so adaptivity is allowed. $\square$

Worked example: two repeated keys. Suppose the cache contains N_+ copies of $\mathbf{1}$ with value $+1$ and N_- copies of $-\mathbf{1}$ with value -1 . Put $T = N_+ + N_-$ and $M = n^{-1} \sum_i q_i$. Its attention mean is

$$u(q) = \frac{N_+ e^{\beta M} - N_- e^{-\beta M}}{N_+ e^{\beta M} + N_- e^{-\beta M}}.$$

The index has at most three occupied cells, at most two in the second chart. An append changes one count; a fresh query requires its scalar projection and two score weights. Theorem 4.2 gives expected query cost $O((n + \log(T+2))B^4)$, even at $\beta = \sqrt{n}$. For a balanced cache, $u(q) = \tanh(\beta M)$. Prefill removes the context-length scan while retaining the work needed to read a fresh query.

5 Finite moments with certified error

The grid uses finite supplied keys. Neural keys are usually irrational, and their required accuracy changes over time. For the complete decoder we maintain a polynomial evaluator at each chosen precision. Taylor feature sums for causal attention are established representations [12, 14]. The following lemma records the finite arithmetic and error guarantee that exact sampling needs.

Lemma 5.1 (A finite moment evaluator). *Let $z_j \in [-1, 1]^r$ and $v_j \in [-1, 1]^a$ lie on a common dyadic grid 2^{-P} . For $\|t\|_1 \leq H$, $H \geq 1$ an integer, an appendable index encloses every coordinate of*

$$A(t) = \frac{\sum_{j \leq T} e^{\langle t, z_j \rangle} v_j}{\sum_{j \leq T} e^{\langle t, z_j \rangle}}$$

to width 2^{-p} . Its finite arithmetic is polynomial in a, H, P, p , rational input lengths, $\log(T+2)$, and, for coefficients with $1/\sqrt{n}$, $\log(n+2)$, at fixed r . The coefficients of t may be rational with a common denominator, or rational multiples of $1/\sqrt{n}$. The index has $(a+1)N_m$ integer counters, where

$$m = \lceil 8(p + 3H + 7) \rceil, \quad N_m = \binom{m+r}{r}. \quad (5.1)$$

Proof idea. Store sums of every monomial up to degree m . The sums are exact; only a finite Taylor truncation and the final certified division need error allowances.

Proof. Put $q = p + 2H + 6$ and $E_m(x) = \sum_{d=0}^m x^d/d!$. The factorial inequality $N! \geq (N/e)^N$, obtained by integrating $\log x$ below its sum, gives

$$\sup_{|x| \leq H} |e^x - E_m(x)| \leq e^H H^{m+1}/(m+1)! \leq 2^{2H-m-1} \leq 2^{-q}.$$

Here $m+1 \geq 8H$ and $e < 4$. Store $S_\nu = \sum_j z_j^\nu$ and $M_{\nu,i} = \sum_j z_j^\nu v_{j,i}$ for $|\nu| \leq m$ as integer numerators over $2^{P(m+1)}$. Each numerator has $O(P(m+1) + \log(T+2))$ bits. Insertion adds exact integer increments, so the index acquires no rounding error with age. The identity

$$E_m(\langle t, z \rangle) = \sum_{|\nu| \leq m} \frac{t^\nu z^\nu}{\nu!}$$

evaluates the truncated numerator and denominator from the counters. If d_t is a common rational denominator for t , the coefficient denominators divide $d_t^m m!$, since $\nu!$ divides $m!$. All intermediate integer lengths are polynomial in the parameters.

Divide both sums by T . The true denominator Z obeys $Z \geq e^{-H}$, the numerator satisfies $|N_i| \leq Z$, and each Taylor error is at most $\delta = 2^{-q}$. Thus the truncated denominator is positive and

$$\left| \frac{\tilde{N}_i}{\tilde{Z}} - \frac{N_i}{Z} \right| \leq \frac{2\delta}{Z - \delta} \leq 4\delta e^H \leq 2^{-p-4}.$$

Rational division to width 2^{-p-2} , enlarged by this error, gives the required interval. If $t_i = c_i/\sqrt{n}$, split monomials by parity of $|\nu|$: each sum is $A + B/\sqrt{n}$ with rational A, B . A common denominator divides $d_c^m n^{\lfloor m/2 \rfloor} m!$. Certified square-root bisection and the lower bound $\tilde{Z} \geq e^{-H}/2$ give the ratio with only polynomial additional precision. This treats $\sqrt{n}$ exactly. $\square$

6 Errors propagate through depth

The maintained evaluator must enclose the network on the actual token history. It cannot treat independently rounded keys as exact model activations. We next construct one deterministic evaluator for every requested precision.

Lemma 6.1 (Uniform prefix evaluation). *For the model of Theorem 2.1, all next-token cumulative probabilities can be enclosed to width 2^{-p} by a maintained causal evaluator. Its per-token work is polynomial at fixed r in $n, D, V, b, s, \beta, p, \log(T+2)$. One may use working precision*

$$P = p + O((D+1)[b + \log(n + s + \beta + 2)] + \log(D + V + 2)). \quad (6.1)$$

A complete replay costs T times the same polynomial.

Proof idea. Attention is uniformly Lipschitz in the largest error over the prefix. Each layer reads the preceding layer's states, so errors accumulate with depth. Exact moment sums prevent an additional recurrence in token count.

Proof. Write $p(a)$ for softmax. Its directional derivative is $\dot{p}_j = p_j(\dot{a}_j - \sum_i p_i \dot{a}_i)$; hence $\|p(a) - p(a')\|_1 \leq 2\|a - a'\|_\infty$, independently of the number of positions. If values are bounded by K , then

$$\|A(a, v) - A(a', v')\|_\infty \leq \|v - v'\|_\infty + 2K\|a - a'\|_\infty. \quad (6.2)$$

For queries and keys bounded by K , their score perturbation is at most $\beta K(\delta_q + \delta_k)$, by subtracting one factor at a time.

Normalization satisfies $\|N_\varepsilon(h)\|_\infty \leq \sqrt{n}$. Its Jacobian has entries $\delta_{ij}/\sqrt{v} - h_i h_j/(nv^{3/2})$, with $v = \varepsilon + \|h\|_2^2/n$; the maximum absolute row sum is at most $(1 + \sqrt{n})/\sqrt{\varepsilon}$. Choose a power of two $K \geq s(\lceil \sqrt{n} \rceil + 1)$ less than twice this bound. All query, key, and value coordinates are then bounded by K . Every affine map has Lipschitz constant at most s , and $\tanh$ at most one. Residual addition, the affine head, and cumulative softmax have polynomial Lipschitz bounds. Thus the whole prefix computation has $J = O(D + 1)$ stages on prefix arrays, each bounded by a common constant $L_0 \geq 2$ with $\log_2 L_0 = O(b + \log(n + s + \beta + 2))$.

If local error at every stage is at most δ , induction gives final error at most $JL_0^J \delta$. Take $P \geq p + \lceil \log_2(J + 1) \rceil + J \lceil \log_2 L_0 \rceil + 10$ and $\delta = 2^{-P}$. A certified midpoint, followed by dyadic rounding, implements a local approximation without deciding whether an unknown real lies exactly on a grid boundary. Compute normalization to its local error allowance, round it to the common grid, and clip to $[-\lceil \sqrt{n} \rceil, \lceil \sqrt{n} \rceil]$. Compute $\tilde{k} = W_K \tilde{N} + b_K$ *exactly* on the resulting grid. It remains in the affine key space. Independent rounding of key coordinates would not preserve that promise. Compute the query and value products similarly; round completed residual updates.

Apply Lemma 4.1 to a column basis of W_K . The common affine key space has coordinates $k = c_0 + Ak_I$ with $|A_{ij}| \leq 2$. Put $z_j = (k_j)_I/K$. The common score offset cancels, and the remaining coefficients are

$$t = \frac{\beta K}{n} q^T A, \quad \|t\|_1 \leq 2r\beta K^2.$$

Take an integer $H \geq \max(1, 2r\beta K^2)$ and apply Lemma 5.1, with the additional $O(\log(K + 2))$ local bits needed to rescale values. The chart has a common determinant denominator of polynomial encoding length. Standard scaling has exactly the coefficient form allowed by that lemma. All moment counters are exact on the common grid for this precision.

Elementary finite series and bisection evaluate $\tanh$ and normalization at dyadic arguments in polynomial time in the requested precision and the stated range bounds. The residual range is at most $s + D(sK + s + 1)$; the affine logit range is consequently polynomial. To evaluate softmax without an exponential range factor, first enclose each logit to width $1/4$ and take A as the largest upper endpoint. Then $z_i - A \leq 0$ for all i , and some $z_i - A \geq -1/4$, so $\sum_i e^{z_i - A} > 1/2$. Intersect refined logit intervals with their original coarse enclosures; the shifted upper endpoints remain nonpositive. Refining the negative exponentials to $2^{-p - O(\log V)}$ and dividing certified intervals gives each cumulative probability to width 2^{-p} . No exact comparison between real logits is required. All local error allowances can be enlarged by fixed guard margins and propagated explicitly.

Process positions in increasing order. At position t and each layer, form its query, key, and value from the preceding-layer state, append that record before the attention query, then complete the residual and feedforward updates. This includes exactly $j \leq t$. In the error induction, take the maximum over every position in a prefix at each layer. A previous token's error is never fed into the same layer's residual state; attention reads the preceding layer. Equation (6.2) is uniform in prefix length, and exact moment additions add no time error. This proves both the certificate and its independence from a linear factor in T . $\square$

For reference, elementary schoolbook arithmetic in the preceding construction permits a deliberately coarse per-token bound

$$\begin{aligned} C_r(D + 1)(n + V + 1)^2(m + 1)^{2r+4}L^4, \\ L = C_r\{m[P + b + \log(n + m + D + V + 2)] + \log(T + 2) + 1\}. \end{aligned} \tag{6.3}$$

Here $m = O_r(1 + \beta K^2 + P)$, and chart lengths are included in b up to the displayed rank-dependent constant. This formula makes the rank dependence visible. The exact feature count (5.1) is more informative for a feasibility check than this loose upper bound.

7 Exact decisions while precision grows

Interval sampling is classical [4, 10]. The issue for a growing neural cache is paying for increased precision. The next lemma separates that scheduling question from the particular network evaluator. Figure 2 outlines the schedule.

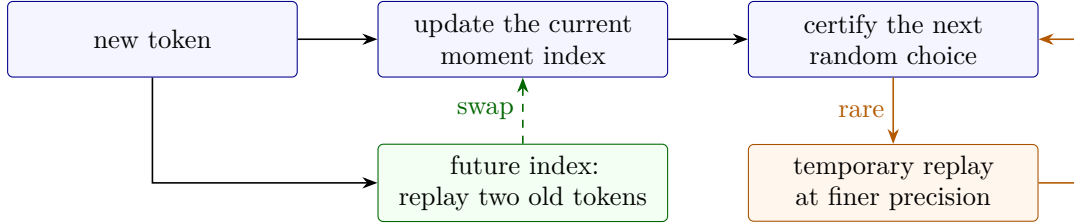


Figure 2: Maintaining an exact decoder. A routine step evaluates the new token in the current index and replays at most two old tokens in a future index. The future index takes over when it catches up. If the random choice is too close to a probability boundary, a temporary finer replay resolves it. All three computations use the same discrete token history.

Lemma 7.1 (Maintained evaluation implies exact sampling). *Suppose a deterministic causal evaluator at accuracy p appends one actual history token and returns certified cumulative probabilities of width 2^{-p} in at most $G(p + \log(T + 2))$ bit operations, where G is a fixed polynomial with nonnegative coefficients. Include in G the work of comparing a p -bit uniform cell with those cumulative intervals; a scan costs $O(V(p + 1))$ bits and can be added to G if needed. Suppose replaying a length- T history has cost at most $TG(p + \log(T + 2))$. Then an exact autoregressive sampler has polynomial expected per-token work after a prompt pass. If $G(u) \leq Au^k$ for $u \geq 1$, put $S = 1 + \log(V + 2) + \log(T + 2)$. Its expected total bit work is $O_k(AS^k + S)$. The expectation conditions on every realized history.*

Proof idea. Use enough routine bits that an ambiguous output decision is rare even after charging a complete replay. Prepare the next routine precision by processing two historical tokens after each append.

Proof. At a power-of-two horizon R , use $p_R = \lceil \log_2(64V(R + 2)^3) \rceil$. Reveal the first p_R bits of a fresh uniform binary expansion U , giving a dyadic cell $[a, b]$. Let $[\underline{F}_i, \overline{F}_i]$ enclose each cumulative boundary, using exact $F_0 = 0, F_V = 1$. Output i if $\overline{F}_{i-1} \leq a$ and $b \leq \underline{F}_i$. If undecided, replay the actual history at accuracy $p_R + 1$, reveal the next uniform bit, and try again; continue similarly. Different replay intervals need not be nested, because all contain the same exact boundaries.

If a precision- p comparison is undecided, U lies within 2^{1-p} of one of at most $V - 1$ boundaries. Conditional on the history,

$$\mathbb{P}(\text{undecided at precision } p) \leq 4V2^{-p}. \quad (7.1)$$

Outside the finite set of true boundaries, the dyadic cell and enclosing errors eventually fit within one categorical interval. The procedure therefore stops almost surely and gives exactly the inverse-

CDF law. The expected temporary replay work is at most

$$8VT2^{-p_R} \sum_{j \geq 0} 2^{-j} G(p_R + j + \log(T + 2)).$$

For $T \leq R$, the prefactor is $O((R + 2)^{-2})$. For $P \geq 1$,

$$\sum_{j \geq 0} 2^{-j} (P + j)^k \leq P^k \sum_{j \geq 0} 2^{-j} (j + 1)^k \leq 2^{k+1} k! P^k.$$

The last inequality follows from $(j + 1)^k \leq k! \binom{j+k}{k}$ and the binomial generating function. Summing costs on the events that reach each precision proves finite expected work without assuming finite expected stopping time first. The initial uniform cell costs $p_R = O(S)$ random bits; the same tail charges all additional random bits.

Let N be the least power of two at least T_0 . During prefill build the routine evaluator at p_{2N} . Use it alone until length N . At length N , start an empty future evaluator at p_{4N} . After each of the next N appends, update the current evaluator once and replay two historical tokens into the future evaluator. After j appends, the true history has length $N + j$ and the future evaluator has processed $2j \leq N + j$ tokens. At length $2N$ it is current. Switch, double N , and start the next future evaluator. Thus every append performs at most three deterministic token evaluations. The future accuracy exceeds the current one by only a constant number of bits. There is no deterministic rebuilding burst.

Retain the actual token history for all replay operations. Temporary fine indices are discarded after the decision; the routine and future indices follow only the fixed deterministic schedule. Each precision computes its own states from the same tokens, without reusing another precision's approximate states. Conditional on any previous history, all intervals therefore refer to the same true next-token law. Fresh uniform bits give that conditional law, and induction gives the whole autoregressive joint law. The construction stores the token history, at most two routine indices, and a temporary index when refinement requires one. The geometric estimate also bounds expected temporary space. $\square$

Proof idea for Theorem 2.1. The finite moment evaluator gives a polynomial routine cost. The replay lemma retains that polynomial when the output decision must be exact.

Proof of Theorem 2.1. Use Lemma 6.1 with the exact counters of Lemma 5.1, then apply Lemma 7.1. Its geometric moment bound preserves the degree in routine precision. Since $K = O(s\sqrt{n})$, the Taylor parameter satisfies $H = O_r(1 + \beta s^2 n)$, which is polynomial at $\beta = \sqrt{n}$. A small positive stabilizer increases the bit guard through $\log(1/\varepsilon) \leq b$, without an exponential dependence in the Taylor degree. The displayed construction uses only finite integer arithmetic, certified series, square-root bisection, and random bits. It includes both routine neural record creation and every replay, which proves the theorem. $\square$

Corollary 7.2 (A fixed finite model). *Fix all parameters of the basic model, including width and key ranks. The expected per-token cost is $O_{\text{model}}(\log^{2r+12}(T + 2))$. A fixed model with full key rank is included, with $r = n$.*

Proof idea. Only precision and integer counter length grow with context length.

Proof. In (6.3), fixed model parameters give $P, m = O_{\text{model}}(\log(T + 2))$ and $L = O_{\text{model}}(\log^2(T + 2))$. Substitution gives the exponent; Lemma 7.1 preserves it. $\square$

8 Quantities to check on a model

The current bounds become unusable as rank grows. The following quantities make that limitation testable before an implementation is attempted.

Let n be residual width and d, a one head's key and value dimensions. Standard scores are $\langle q, k \rangle / \sqrt{d}$; RMSNorm averages over n . Fold its learned gain into every following map: $W^{\text{eff}} = W \text{diag}(g)$. The effective row budget is $\max_i \sum_j |W_{ij} g_j|$.

Certificate	Cost quantity to evaluate
Common affine key rank r and coordinate bounds $Q_{\max}, K$	$M_T \leq \min\{T, (r+1)(5 + 32r\bar{\beta}Q_{\max}K)^r\}$; index query work $O_r((d + M_T + \log T)B^4)$.
An exact score sign at radius A_{coin}	Unindexed attention uses at most $2e^{2A_{\text{coin}}} - 1$ predecessor requests.
Weighted tanh row radius $\rho = \sum_j w_j R_j$ and row bias $ c \leq 1$	At most $2\rho(1 + \rho)$ source requests.
RMS floor $\lambda \leq \varepsilon + \ h\ _2^2/n$ at radius R	$\kappa = \max\{1, R^2/\lambda\}$; the floor factory uses $O(1 + \kappa^2)$ source requests at radius $2R/\sqrt{\lambda}$.

The first row is Theorem 4.2 for keys of dimension d . The other rows, the decoder for rectangular heads, and the claims of the next paragraph are proved in the full version [17].

The rank certificate includes every allowed position. Rotary maps can enlarge the common key span; a small rank for each individual transformed matrix does not bound their joint span. Even a rank-one unrotated image can have a full common rotated span. For unrotated heads, the exact score rank $\text{rank}[W_Q^T W_K; b_Q^T W_K]$ can replace key rank after a proved recoding. An approximate-space supplied-cache index is also exact when its certified coordinate residual η satisfies $\bar{\beta}Q_{\max}\eta \leq 1/8$. Its acceptance test uses the actual score. This is a finite-cache result, with no approximate-rank extension claimed for the neural moment decoder.

For a bounded chart, the full decoder uses $H = \lceil \max\{1, 2r|\beta|Q_{\max}K\} \rceil$ as a safe coefficient bound. At attention precision p it stores

$$N_m = \binom{m+r}{r}, \quad m = \lceil 8(p + 3H + 7) \rceil$$

features, with $(a+1)N_m$ counters per head and precision index. The routine precision is $O(\log T)$ plus a stability guard computed from row norms and normalization floors. The full version gives the token work polynomial and a numerical worksheet for these counts. It retains its rank-dependent constants. Even at illustrative degree $m = 256$, rank eight requires more than $5 \cdot 10^{14}$ features.

Weights give exact row sums and ranks. A score maximum or energy minimum observed on selected texts describes those contexts; a universal theorem needs a certificate for every allowed history. Positive inside-root ε always gives $\lambda = \varepsilon$. For additive residuals, record each update radius and use the product in the additive theorem, including the final head. A small measured score range need not supply the score-sign interface required by the Poisson factory. Verification that reads the current context is part of online work. The formulas predict construction sizes and bit-work bounds; timing and speedup require calibrated implementations.

9 The critical exponent and the limits of the results

For the scalar depth question, consider $h_0 = x \in \{-1, 1\}^n$ and $h_\ell = \tanh(W_\ell h_{\ell-1})$ with $s(W_\ell) \leq 1$. The target sign has mean $h_{D,1}$; only weights may be preprocessed. The general exact sampler in

the full version uses $O(D + 1)$ expected input requests and $O(B^4(D + 1))$ bits. Worst-case lower bounds have order $\sqrt{D}$. For every fixed first-layer rank cap r , the upper bound becomes $O_r(\sqrt{D + 1})$ requests and $O_r(B^{\max\{4, r+2\}}\sqrt{D + 1})$ bits, within the original preprocessing budget $O_r(Dn^2 \log^6 n)$ at logarithmic depth and $b = O(\log n)$. Matching lower statements require $n \geq D + 1$ and enough coefficient precision. Rectangular batching [16] supplies the last preprocessing step.

At arbitrary rank, put $C_{\text{col}} = \sum_j \max_i |w_{1,ij}|$ and $K_{\text{col}} = \max\{1, C_{\text{col}}\}$. The full version proves $O(K_{\text{col}}^2 \sqrt{D + 1})$ probes and $O(K_{\text{col}}^7 \sqrt{D + 1} B^{40})$ bits under the same budget. One probe represents an implicit vector; sampled derivatives along at most four network paths implement its empirical correction. A Hadamard matrix divided by its width has full rank and $C_{\text{col}} = 1$. The envelope can also be n .

Main open problem. Does every unrestricted zero-bias, row-norm-one network admit an exact sampler with $O(\sqrt{D + 1})$ expected input probes under the same weight preprocessing budget, and with only polylogarithmic bit overhead? A bound of $\sqrt{D + 1} \log^{O(1)}(D + 2)$ probes would already close the depth exponent. The current worst-case probe exponent remains between $1/2$ and 1 . Logarithmic factors in width must not be hidden in this query claim: at $D = \lceil \log_2 n \rceil$ they could absorb the entire gap. Fixed rank and bounded column envelope leave the general class uncovered.

Two method-specific limits help explain the gap. A stopped-score inequality forces every normalized scalar factory for $\tanh(\rho z)/\tanh \rho$, including bounded real estimators, to use at least $1 + \rho^2 + O(\rho^4)$ expected requests. Grouped matrix second-order score witnesses are at most $\sqrt{6D}$. The full version proves both limits; neither settles global query complexity.

For arbitrary fresh queries against a prefill index, Haverbeck et al. prove a conditional attention barrier after polynomial preprocessing [11]. The full version gives a bounded-coordinate, standard-scaling, binary-token reduction under randomized Orthogonal Vectors Hypothesis. At width $O(\log^2 T)$ it excludes a uniform $T^{1-\epsilon} \text{poly}(n, B)$ query algorithm for any fixed $\epsilon > 0$ and any fixed polynomial prefill budget. The reduction uses a constant final-token probability gap. Earlier high-accuracy attention and KDE hardness theorems do not alone supply that sampling conclusion [1–3]. The fixed-rank theorem does not remove this growing-dimensional barrier.

Relation to approximation. KDEformer approximates an attention matrix product in operator norm; HyperAttention gives an operator-norm guarantee under heavy-entry and column-norm conditions; MagicPIG uses a prefill-built LSH index and importance-weighted ratio estimates during decoding [5, 9, 20]. These approximation guarantees do not certify exact next-token laws. Maintaining causal features and refining probability intervals are prior techniques; our finite arithmetic, depth-uniform error control, and maintenance bound support the exact law. Analytic factories and the scalar debiased construction [15] leave the unrestricted multivariate coefficient problem open.

Evidence and formal status. No trained checkpoint or sampler runtime was measured. Numerical and exact-arithmetic checks in the source supplement the proofs and are labeled separately. A Lean 4 formalization with Mathlib, in the repository <https://github.com/SamMausberg/exact-sampling-networks>, checks the exactness statements, probability laws and analytic inequalities in the proofs of Sections 3–7 using only the standard axioms; some steps enter as explicit hypotheses. The bit-cost accounting and the certified arithmetic routines are not formalized.

AI disclosure. GPT-6 Astra Pro and Claude models assisted with the mathematical proofs and the Lean formalization.

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